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RAID 1 Server Configuration Documentation

Prepared by the IT Department

Duration : 3 Weeks

May 2026

Documentation Location: \\192.168.0.70\jgc3\JGC3\DEPARTMENT\IT\Documents
Date: 25th May 2026

Phase 1: Hardware Preparation and HDD Installation

The first phase of the RAID 1 server setup focuses on preparing the server chassis and ensuring that the hard disk drives (HDDs) are properly mounted, cleaned, labeled, and connected before configuration begins. Proper preparation is important to reduce hardware failure risks, improve airflow within the chassis, and simplify future maintenance or troubleshooting processes.

Figure 1.1 – New 2TB WD Red HDD's have been screwed into the HDD Tray.

The HDD tray was first removed from the front panel of the server chassis. Before installing the drives, the tray and surrounding internal components were cleaned to remove dust and debris that could affect airflow or damage the hardware over time. Vacuuming and cable tidying were also performed to maintain a cleaner internal environment and improve cooling efficiency.

Each HDD was labeled beforehand to simplify drive identification during RAID configuration and maintenance procedures. The drives were then positioned onto the tray with the SATA ports facing inward toward the case to ensure proper cable routing and accessibility. The HDDs were securely mounted using the provided screws to minimize vibration and maintain drive stability during operation.

Figure 1.2 – The Server PC's front internal view during maintenance.

The server chassis contains multiple drive bays positioned behind the front panel, with each slot assigned a numbered drive position from top to bottom. Proper slot placement was important because each drive corresponds to a specific SATA data cable connected to the motherboard. Correct installation and identification of the drives help prevent configuration errors during the RAID setup process.

After the drives were inserted into their assigned bays and connected properly, the server chassis was closed and prepared for the next phase of logical storage configuration.

Phase 2: RAID 1 Storage Pool and Mirrored Disk Configuration

The second phase focuses on transitioning from physical hardware installation into logical storage configuration by creating a RAID 1 mirrored environment. This phase ensures that both HDDs work together as a redundant storage system capable of protecting data in the event of a drive failure.

Once the server was powered on, the installed drives were verified within the operating system to ensure that both HDDs were detected correctly. The Disk Management utility was then accessed to initialize and prepare the newly installed drives for RAID configuration.

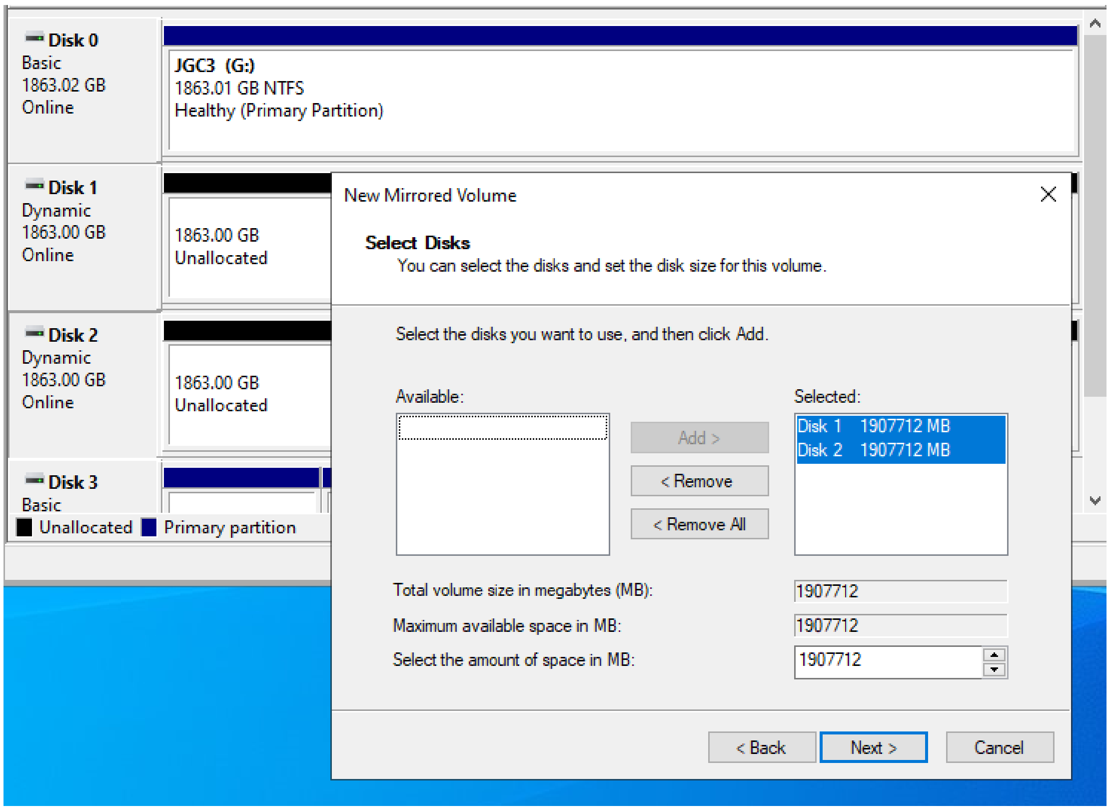

Figure 2.1 – First setup of “New Mirrored Volume” for Disk 1 & 2.

The two HDDs were converted into Dynamic Disks to enable advanced storage features required for RAID 1 mirroring. After initialization, a new mirrored volume was created by combining both drives into a single RAID 1 storage environment. In this setup, all data written onto the primary drive is automatically duplicated onto the secondary drive in real time.

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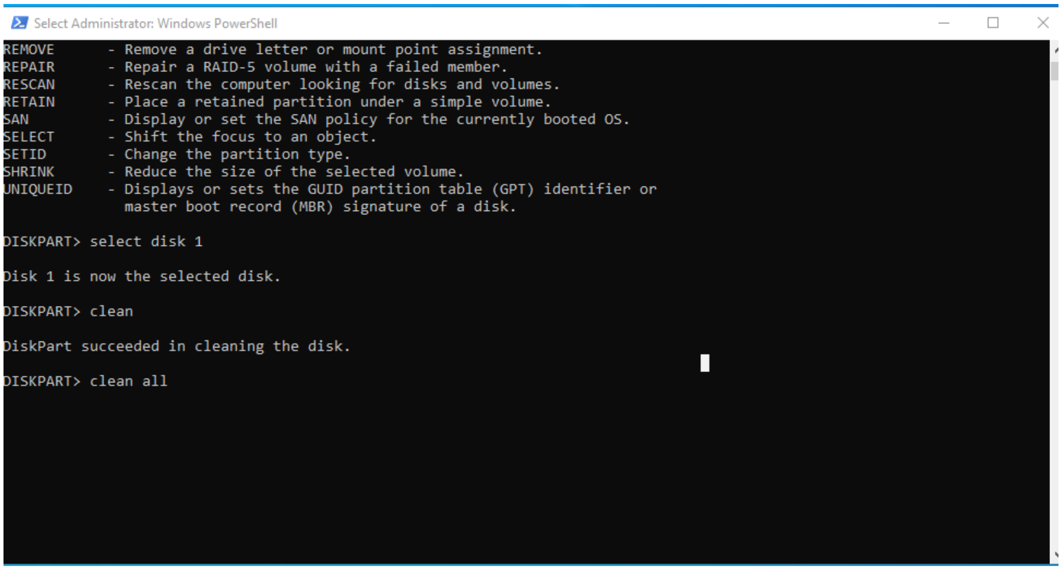

Figure 2.2 – Cleaning the Disks to make sure it's cleanly in a zero-binary state.

The purpose of RAID 1 is to provide fault tolerance and improve data reliability. If one HDD fails, the second mirrored drive will still contain an identical copy of the data, allowing the server to continue operating without immediate data loss. This configuration is especially important for server environments where uptime and file integrity are critical.

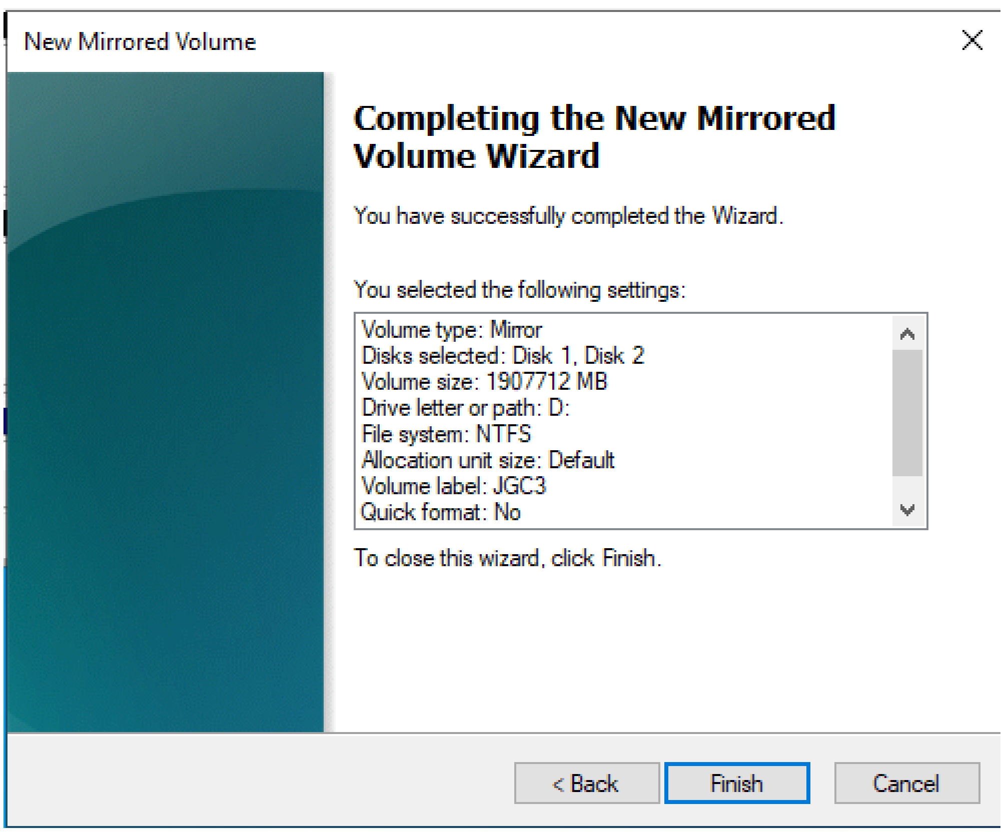

Figure 2.3 – Finishing setup of the “New Mirrored Volume” on Disk Management.

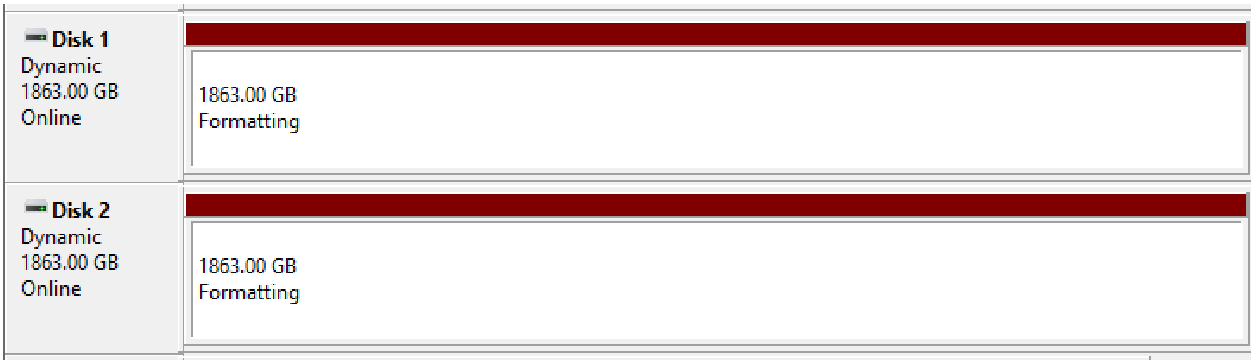

Figure 2.4 – The two Disks undergoing first-time formatting.

Once the mirrored volume was successfully created, the storage pool was formatted, assigned a drive letter, and verified to ensure that both disks were synchronizing correctly before proceeding to data migration.

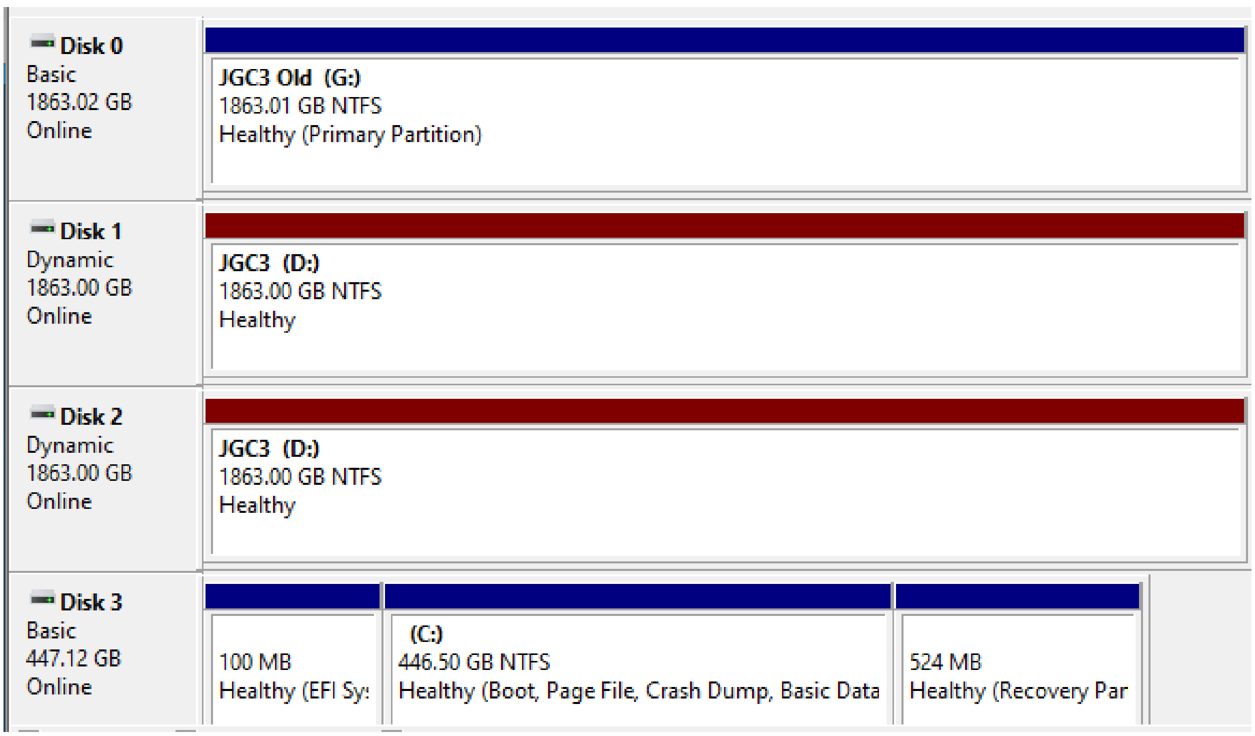

Figure 2.5 – After setting up the RAID 1 – Mirroring for Disk 2 to Disk 1, Disk Management displays “Healthy” Mirrored (Red) Dynamic Disks for Disk 1 and 2.

Phase 3: Migration of Existing Server Files into the RAID 1 Environment

The third phase involves migrating the existing server files from the original solid-state drive (SSD) into the newly created RAID 1 mirrored storage environment. This migration process ensures that operational files, shared folders, and important data are transferred into the redundant storage system for improved reliability and protection.

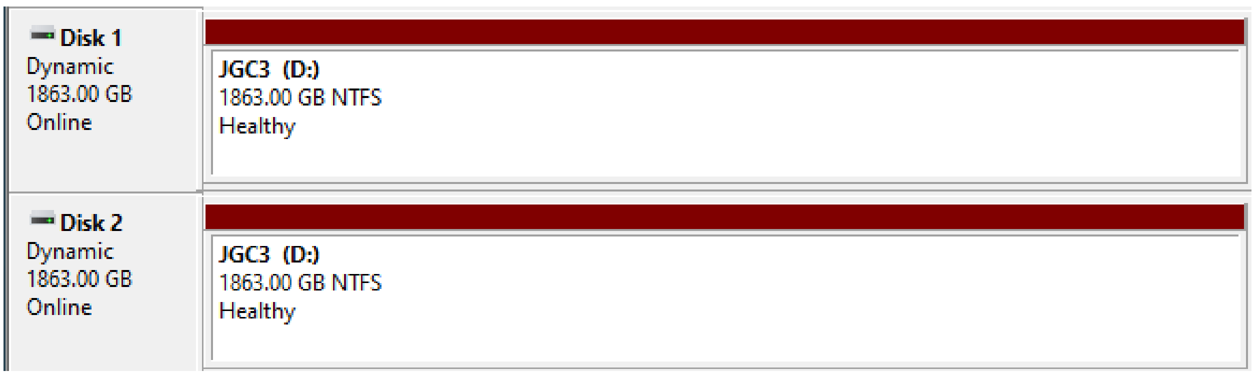

Figure 3.1 – Double checking the 2 disks to be in healthy condition.

Before the migration process began, the RAID 1 mirrored volume was verified to ensure that both HDDs were functioning properly and synchronizing correctly. Folder structures and storage directories were prepared within the new mirrored drive to maintain organized file management during the transfer process.

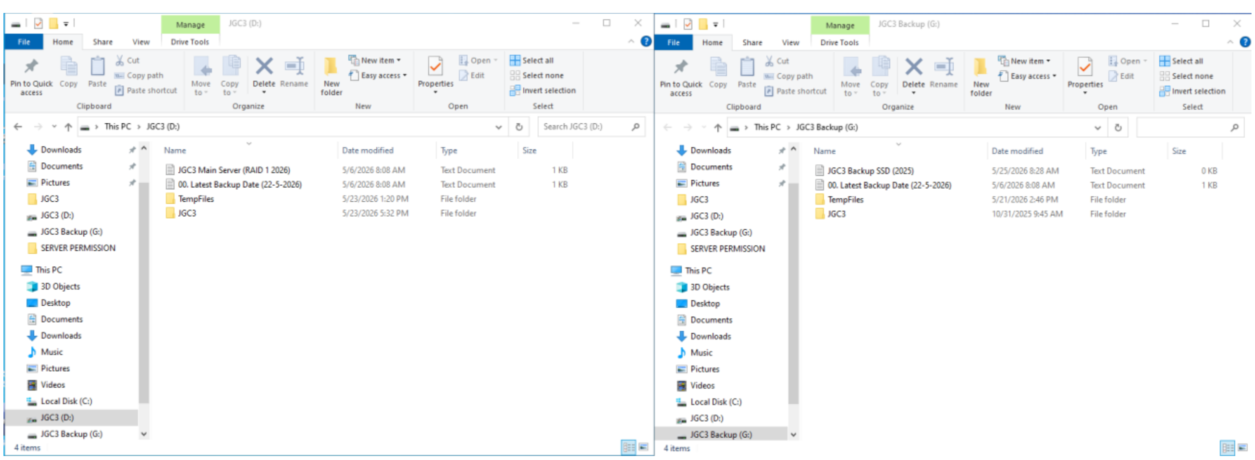

Figure 3.2 – Migration Complete from Old JGC3 (Now JGC3 Backup) into the JGC3 Main.

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Existing server files from the SSD were then copied into the RAID 1 volume. The migration process included validating that all critical files, directories, and permissions were transferred successfully without corruption or missing data. This step was important to ensure that the server environment could continue operating normally after migration.

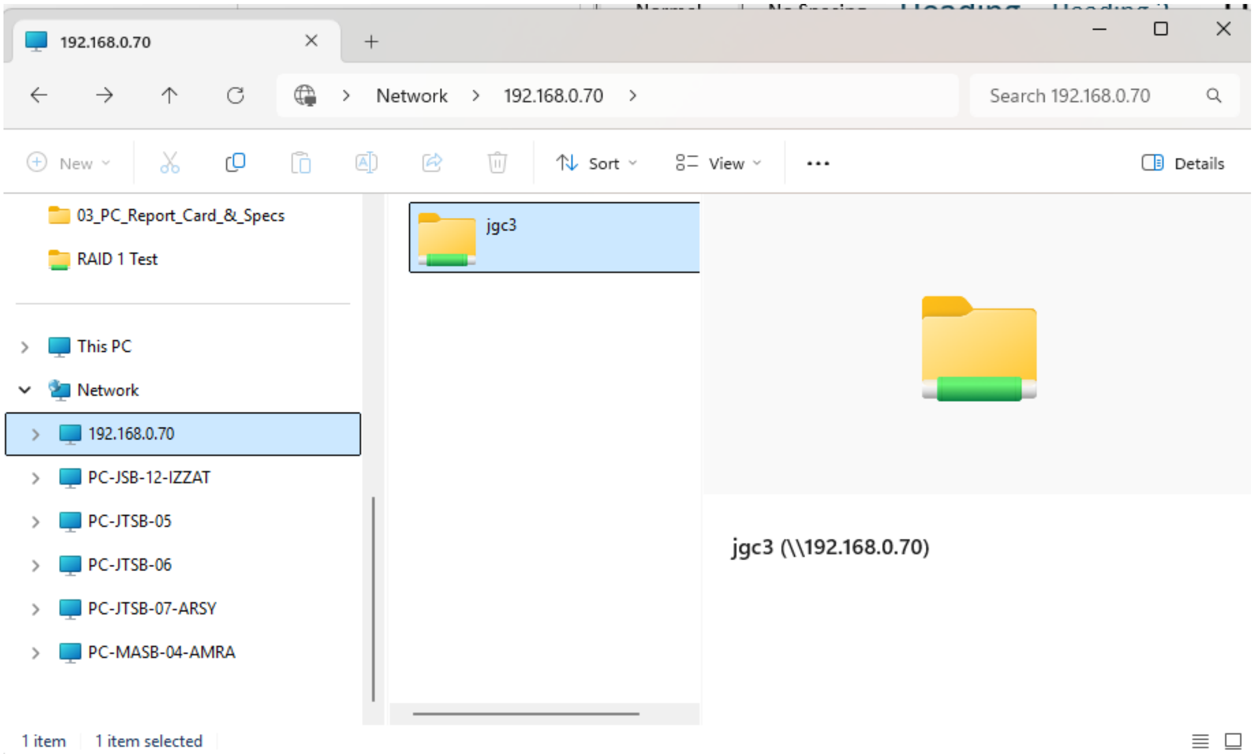

Figure 3.3 – The JGC3 can be accessed at another PC.

After the transfer was completed, several verification checks were performed by comparing files between the original SSD and the new mirrored storage. This was done to confirm data integrity and ensure that the RAID 1 environment was fully operational before transitioning the server workload onto the new storage system.

The successful migration of server files into the RAID 1 environment significantly improves data redundancy and reduces the risk of permanent data loss caused by hardware failure.

Phase 4: Server Sharing and Security Permission Configuration

The fourth phase focuses on configuring both sharing and security permissions for the server environment to ensure controlled access to files and folders within the RAID 1 storage system.

After the RAID 1 storage volume was prepared and the files were migrated successfully, shared folders were configured to allow authorized users to access server resources through the local network. Folder sharing settings were adjusted according to organizational requirements to ensure proper collaboration and accessibility between users and departments.

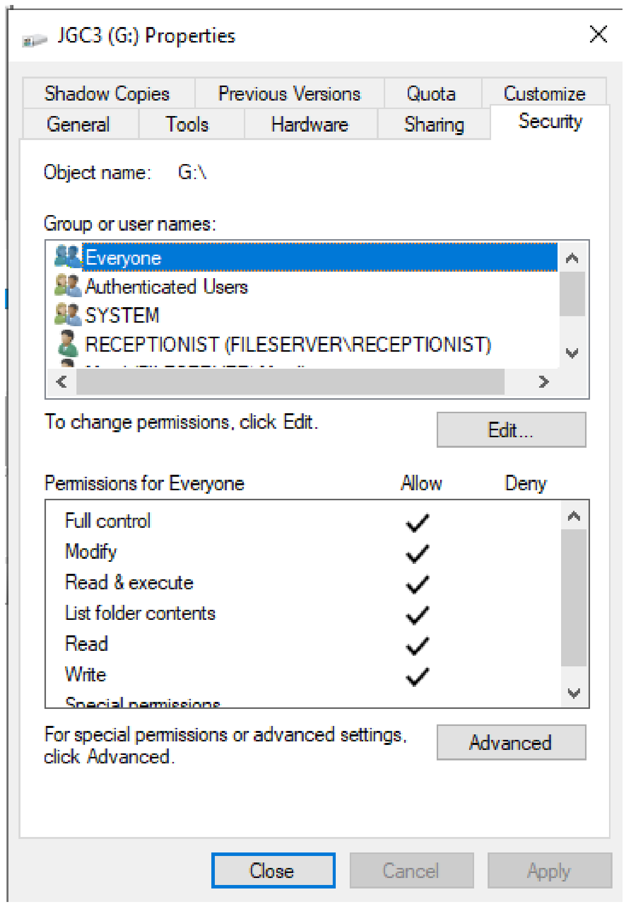

Figure 4.1 – Setting Server Permissions for the new shared JGC3 File.

In addition to sharing permissions, security permissions were also configured using the Windows Security settings. User accounts and groups were assigned different permission levels such as Read, Write, Modify, or Full Control depending on their roles and responsibilities within the server environment.

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The implementation of proper security permissions is important to prevent unauthorized access, accidental deletion, or modification of sensitive files. By separating sharing permissions from security permissions, the server environment maintains a more secure and manageable access control structure.

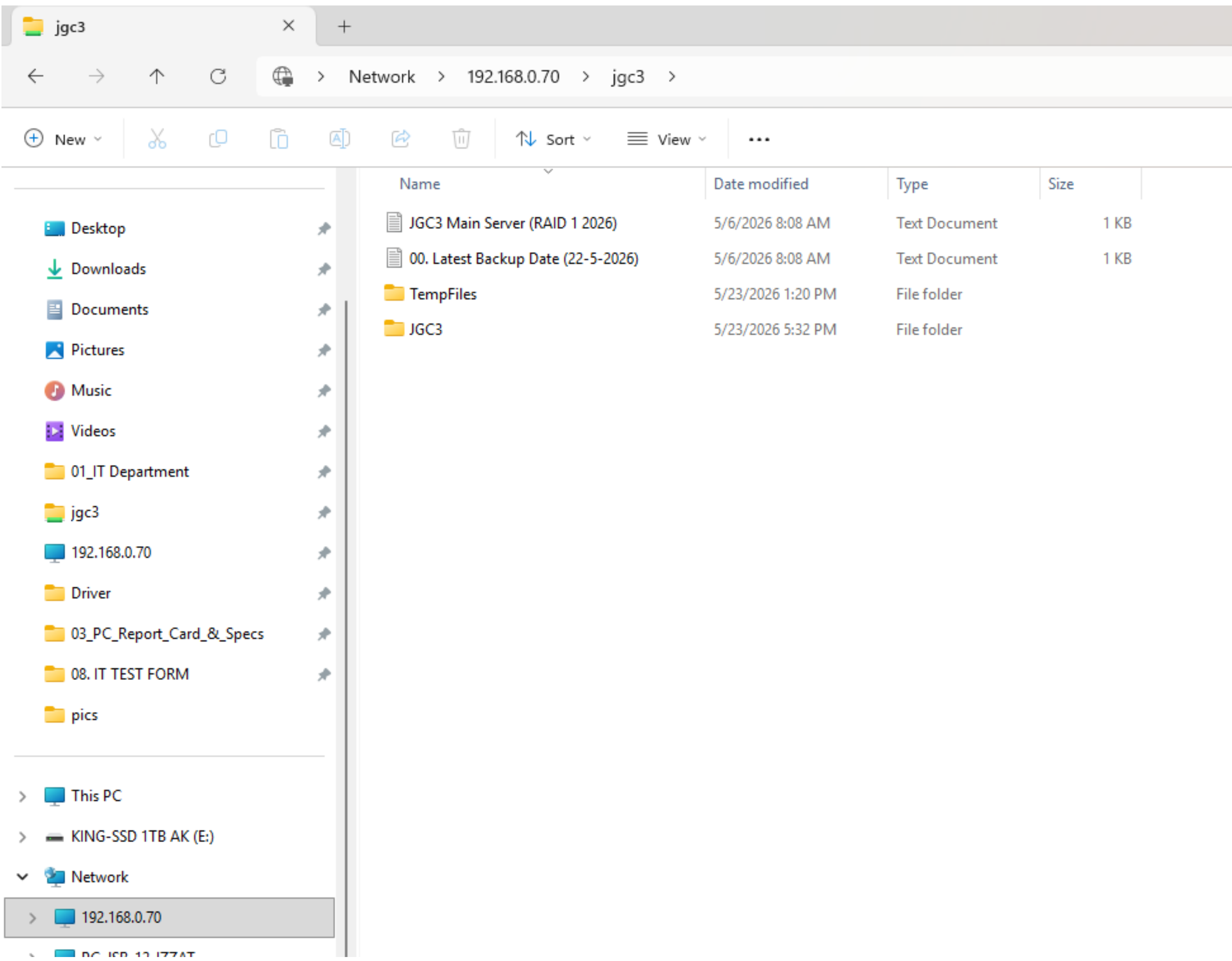

Figure 4.2 – JGC3 can be accessed and used on another computer.

Testing was also performed to verify that users could only access the folders and actions permitted to them. This ensured that the configured permissions operated correctly and securely before the server was deployed for regular use.

Phase 5: SSD Backup Configuration and Recovery Preparation

The fifth phase focuses on repurposing the original SSD as an automatic backup solution for the RAID 1 server environment. While RAID 1 provides redundancy against drive failure, it does not replace the need for backup systems in situations such as accidental deletion, corruption, ransomware, or human error.

The SSD was configured as a dedicated backup destination for critical server files stored within the RAID 1 mirrored environment. Automated backup tasks were prepared to periodically copy important files and system data from the RAID storage into the SSD.

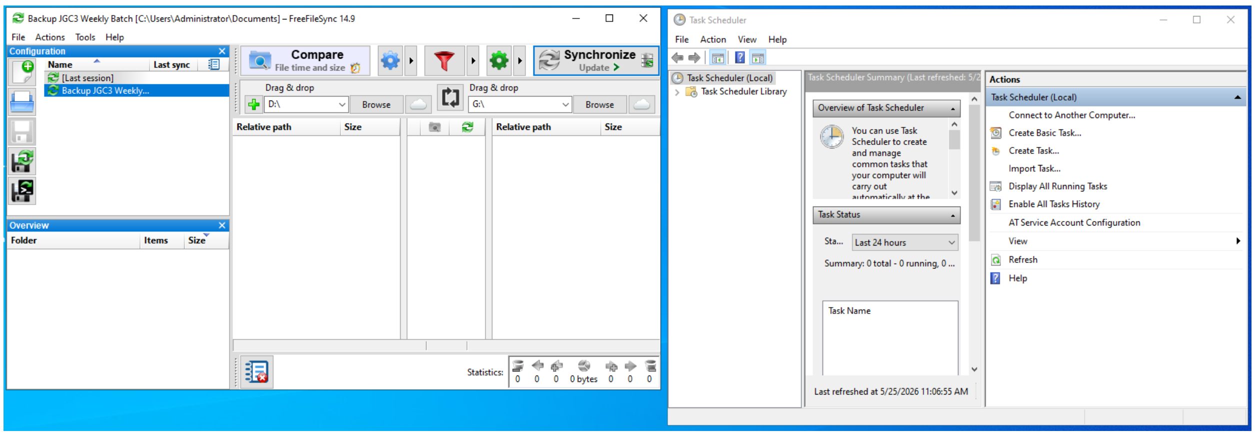

Figure 5.1 – FreeFileSync Setup for Automated Weekly Sunday Backups.

This backup process creates an additional recovery layer separate from the mirrored HDDs. In situations where both RAID drives become corrupted or files are mistakenly deleted, the SSD backup can be used to restore previous versions of the data.

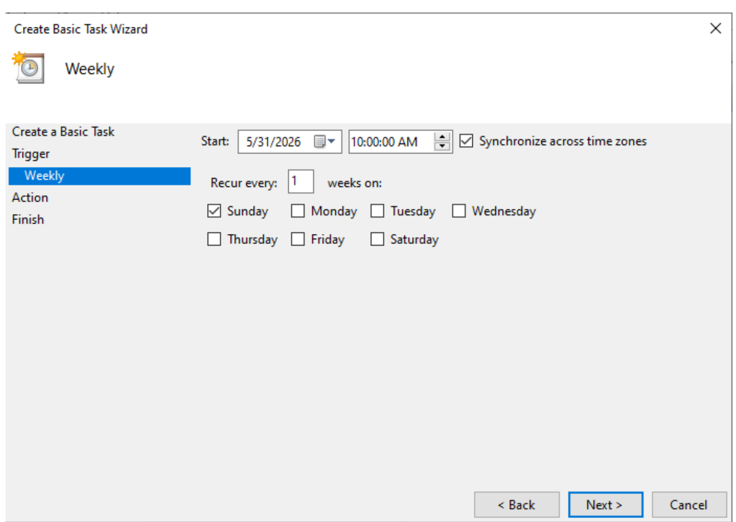

Figure 5.2 – Settings for Basic Task in Task Scheduler.

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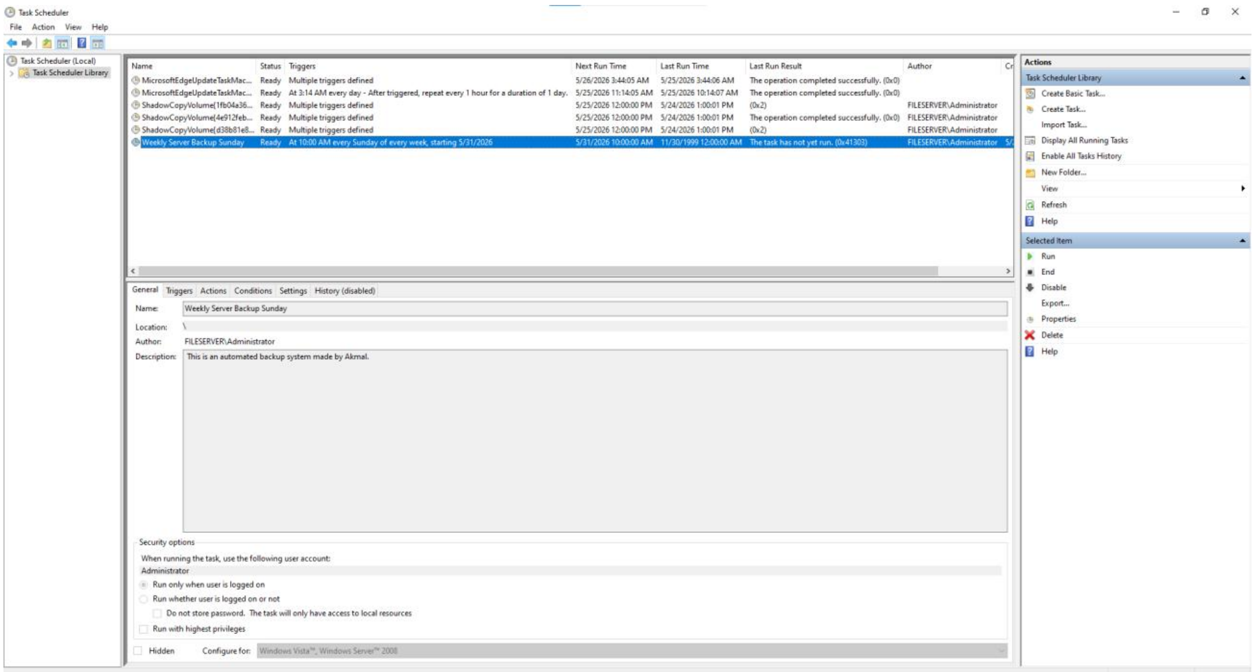

Figure 5.3 – Task Scheduler receiving the task and will backup every Sunday.

Backup scheduling and verification procedures were also considered to ensure that backups remain updated and functional over time. Periodic testing of the backup files is important to confirm that recovery operations can be performed successfully when required.

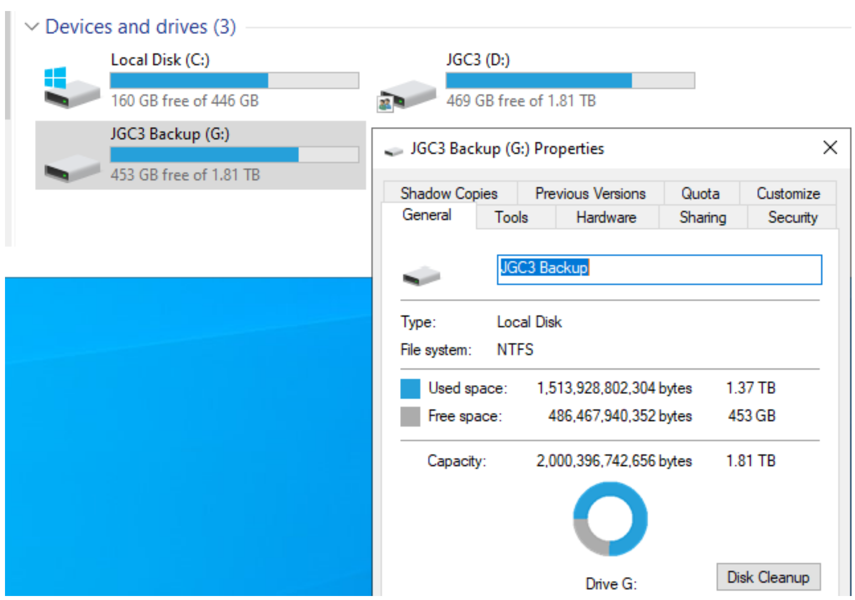

Figure 5.4 – Secondary Backup (Weekly) for the Main Server.

Figure 5.5 – Tertiary Backup (Monthly) for the Server's External HDD.

By using the SSD as an automatic backup solution, the server infrastructure gains improved resilience, recovery capability, and long-term data protection. We also include the External HDD as the Tertiary Monthly Backup (As a safety fallback feature).

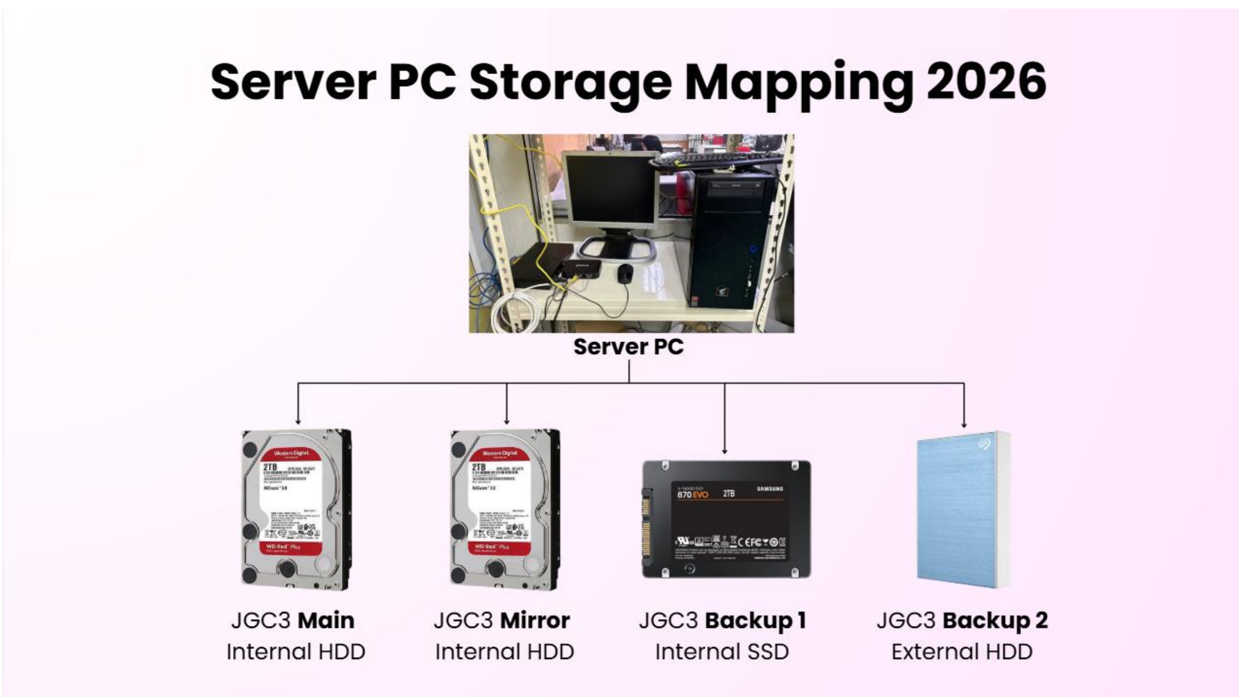

Figure 5.6 – Server PC Storage Mapping 2026.

Final Thoughts and Future Improvements

The RAID 1 server setup successfully establishes a more reliable and fault-tolerant storage environment by combining mirrored HDD redundancy with additional SSD backup protection. Through proper hardware preparation, logical RAID configuration, controlled permission management, and backup implementation, the server environment is better prepared for long-term operational stability and data protection.

The implementation of RAID 1 significantly reduces the risk of downtime caused by HDD failure, while the SSD backup solution provides an additional recovery layer beyond standard mirroring capabilities. The overall setup also improves storage organization, maintainability, and future scalability for server operations.

For future improvements, several enhancements may be considered to further strengthen the server environment. These include implementing automated backup scheduling software, adding cloud backup integration, monitoring HDD health using SMART diagnostics, and configuring notification systems for hardware failures or synchronization issues.

Figure 6.1 – Uninterruptible Power Supply (UPS).

Additional improvements such as upgrading network infrastructure, introducing UPS power protection, and implementing centralized server monitoring tools may also help improve server performance, reliability, and disaster recovery readiness in the future.

ALL DRIVES	PLAN
Internal Hard Disk Drive - JGC3 Server  	• Used as the **Main** Mirror RAID System. • **Automatic Synchronization (24/7 Mirroring)**. • 24/7 NAS Server Compatible. • When One HDD is broken (hardware or software), can easily be replaced with the same or upgrade to a bigger size HDD.
Internal Solid State Drive (JGC3 Backup) 	• Used as the **Secondary** Backup. • Automated Backup **Weekly** on Sunday, at 10:00AM. • When it's time to upgrade, can easily change to a bigger storage pool for SSD Automatic Backup. • When both the HDD's are broken or malfunction, this SSD acts as an immediate backup that can be set and share as the server to prevent very long downtimes.
External Hard Disk Drive (SG SERVER) 	• Used as the **Tertiary** Backup. • Manual Backup **Monthly** on last day of the month. • In a disaster case when the server is encountering HEAVY issues, this External HDD will be the disaster recovery key. • Can be replaced with a bigger storage at any given time (Flexible).

Figure 6.2 – All the drives and their planning.

System Improvement Overview			
Category	**Before Improvement**	**After Improvement**	**Business Impact**
Data Protection	Single point of failure	RAID 1 Mirror + Multi-tier backup	Much safer data handling
Backup Method	Mostly manual	Automated weekly backups	Less human error
Recovery Capability	Slow recovery	Faster disaster recovery	Reduced downtime
Storage Reliability	One drive failure risks total outage	HDD redundancy with mirroring	Higher operational stability
Scalability	Difficult to expand	Easy HDD/SSD upgrades	Future-proof infrastructure
Server Availability	Risk of long interruptions	24/7 synchronized operation	Better company productivity
Disaster Recovery	Limited recovery options	SSD + External HDD fallback	Strong business continuity
Maintenance Efficiency	Manual monitoring	Structured backup flow	Easier IT management

Figure 6.3 – System Improvement Overview Table.

Estimated Downtime Reduction		
Situation	**Old System**	**New System**
Single HDD Failure	Server may stop working	Server continues operating
Accidental File Corruption	Difficult recovery	Recover from SSD backup
Major Server Failure	Long downtime	External HDD disaster recovery
Backup Recovery Time	Several hours/days	Faster restoration process

Estimated Result:

- Downtime reduced by approximately **60%–80%**
- Faster file recovery during emergencies
- Better operational continuity for staff

Figure 6.4 - Estimated Downtime Reduction Table.

Company Benefits

Operational Benefits

- Continuous server operation even during HDD failure
- Reduced employee interruptions
- Better file accessibility and reliability

Financial Benefits

- Lower risk of expensive data loss
- Reduced emergency maintenance costs
- Longer hardware usability through structured backups

Technical Benefits

- Easier future upgrades
- Better storage organization
- Improved backup management process

Security & Recovery Benefits

- Multiple recovery points
- Reduced chances of permanent data loss
- Safer business document storage

Figure 6.5 – Company Benefits Table.

List Of Checkers

Prepared By:	Verified By:	Initials By:	Initials By:	Initials By:
……………………..	……………………..	……………………..	…………………….	……………………..
IT EXECUTIVE	HR / PA	DDRA	MRA	TSA

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